

Article

Determining the Applicability and Use of the Intrinsic Motivation Inventory in the Context of Water Conservation

Kristin E. Gibson ^{1,*} , Olivia M. Erskine ² , Kevan W. Lamm ² , Alexa J. Lamm ² , Laura A. Warner ³  and Jessica Holt ² 

¹ School of Natural Resources, University of Missouri, Columbia, MO 65203, USA

² Department of Agricultural Leadership, Education, and Communication, University of Georgia, Athens, GA 30602, USA; oliviaerskine@uga.edu (O.M.E.); kl@uga.edu (K.W.L.); alamm@uga.edu (A.J.L.); jaholt@uga.edu (J.H.)

³ Department of Agricultural Education and Communication, University of Florida, Gainesville, FL 32611, USA; lsanagorski@ufl.edu

* Correspondence: kristingibson@missouri.edu

Abstract: Urgent action is needed to educate the public and encourage water conservation behaviors amidst the global threat of water scarcity impacting billions by 2050. Crafting tailored educational programs about water conservation is crucial for incentivizing behavior change and achieving the Sustainable Development Goals. Intrinsic motivation, which is shaped by personal interests and values, plays a fundamental role in understanding individual behavior in various contexts, and therefore may aid in our understanding of water conservation behaviors. Despite the widespread application of the intrinsic motivation inventory (IMI) to measure intrinsic motivation, its adaptation and utilization in water conservation education remains largely unexplored, prompting this study to establish a relevant scale for intrinsic motivation in the context of water conservation education. Data were collected from 907 respondents in Florida, Georgia, and Alabama and subsequently analyzed using an exploratory factor analysis and confirmatory factor analysis to establish and confirm factor structures related to intrinsic motivation, perceived competence, perceived choice, and relatedness in water conservation. The study findings established the Water Conservation IMI as a reliable tool for measuring intrinsic motivation in water conservation education. Further validation of measures for extrinsic motivation in environmental literature are needed to enhance comprehensive education strategies for sustainability.

Keywords: intrinsic motivation inventory; water; conservation education



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1. Introduction

Water scarcity around the globe may be the biggest threat to humankind in the 21st century [1,2], with more than 5 billion people likely to be water-scarce by 2050 [3]. Around the world, severe droughts, areas experiencing droughts, and water consumption are all increasing due to climate change and population rise [4]. In regions like sub-Saharan Africa, millions of people have been impacted by water scarcity that limits day to day activities [5]. In Central and South America, there are prolonged dry spells that cause major disruptions in the agricultural system [6]. In the southwestern United States (U.S.), stringent water usage policies have been implemented due to the dwindling water supply [7]. The water crises may hinder future food security, economic growth, natural systems, and biodiversity [2].

The United Nations (UN) declared water availability and sustainable management as Goal 6 of their Sustainable Development Goals (SDGs), recognizing the importance of water conservation for healthy ecosystems and human health. However, water is often provided at a low cost for domestic use in the U.S., and many consumers are not aware of the severity of water scarcity [8,9], underscoring the importance of public education

on water consumption. Managing demand-side approaches, like water conservation in the home, and supply-side approaches, like updating water infrastructure, is essential for addressing water scarcity [10].

From a demand-side perspective, encouraging household water conservation is important to the global water crisis [11]. Household water saving is generally achieved through behavior change or water-saving technology adoption [12]. Conservation technology is meant to be a more permanent solution, and current technology has been proven to reduce 35–50% of water consumption in homes [12]. Public policy that supports water-saving technology adoption is needed, especially in support of lower-income household adoption, where individuals may not be able to acquire water-conserving hardware due to financial restrictions [13]. For example, households with a family income of less than USD 24,999 per year may be focused on basic needs and only have the ability to conserve water if the technology is provided for them at no cost [13].

There is also potential for voluntary household water conservation behaviors [14]. However, one of the main barriers is knowledge of the most effective ways to reduce water consumption in the home [15]. Information campaigns have seen small victories in water-saving behaviors [16,17], but more research is necessary to focus on targeting messages for specific audiences [15]. Household water conservation is influenced by several factors, including motivations and abilities, which are crucial to integrate into effective messages to influence behavior [18]. Water conservation educators need access to rich audience data quantifying these factors for their audiences and tools they can use to assess these factors themselves, which currently need to be improved [19].

Current studies in water conservation are shifting towards a demand-side approach, where individual behaviors are targeted [20]. Obringer and White (2022) examined water conservation attitudes in Denver, Las Vegas, and Phoenix (three large U.S.-based cities) to develop archetypes based on cluster values [20]. The archetypes are intended to help aid in developing education strategies for the cities to guide water conservation demand management [20]. Sanchez et al. (2023) conducted a systematic and bibliometric literature review of the literature published from 1984 to 2023 to identify household water-conservation behavior determinants [21]. The authors found that most studies on the factors influencing household water conservation and consumption do not incorporate theories or frameworks. In addition, the most frequently examined variable is attitude. The authors suggest new research goes beyond attitudinal variables to help understand actual behavior [21].

Individual behaviors are complex and situational [22]. Intrinsic motivation is a concept that may help integrate the complexities of audiences into an approach for researchers and practitioners to address public water-saving behaviors. Addo et al. (2018) conducted a meta-analysis of water-use behavior and capability, opportunity, and motivation dimensions [18]. They found that all three dimensions accounted for 37% of the variance in household water use. Steinhorst and Klöckner (2018) examined behavioral information frames with German energy provider clients and found their “results support the mediating role of intrinsic motivation for positive effects of environmentally framed behavioral information on a pro-environmental target behavior” ([23], p. 1021). However, there does not appear to be a consensus in the environmental literature as to how to quantitatively measure intrinsic motivation. Therefore, the purpose of this study was to determine if the intrinsic motivation inventory (IMI) could be adapted to measure water conservation motivation so that the impact of sustainability interventions is quantifiable.

1.1. Motivation for Behavioral Action

Motivation can occur intrinsically and extrinsically. Intrinsic motivation is driven by personal interests or values, whereby individuals act solely for the gratification of the activity [24–26]. Increased interest, excitement, and satisfaction about an action result from acting intrinsically, which later translates into higher performance, persistence, and creativity [26]. Intrinsic motivation is valuable for learning about water conservation because it relates to cognitive processing and achievement [25].

Ryan and Deci (2010) identified three factors influencing individuals to perform actions intrinsically based on the social environment: competence, autonomy, and relatedness [27]. Learning and engagement around a topic may be optimized when competence, autonomy, and relatedness are satisfied [28]. An individual's competence may be enhanced when challenges are achievable, feedback is constructive, and evaluations are not belittling [26]. An individual's autonomy may be enhanced when there is a choice to perform a task and feelings are acknowledged. Some extrinsic rewards can undermine feelings of autonomy, such as imposed goals or demeaning evaluations [26]. An individual's relatedness may be enhanced when appreciation is expressed, or when relationships are secure [26]. Modeling a behavior performed by someone who is relevant and important encourages intrinsic behavior [26]. Although competence, autonomy, and relatedness influence intrinsic motivation in social environments, it is unlikely motivation to engage in a behavior will occur intrinsically unless the individual performing the behavior is interested in performing the behavior [26].

Intrinsic motivation has been studied in the context of sustainability and conservation behaviors. Li and Wen (2019) examined bicycle-sharing users' participation intention in 22 provinces in China [29]. The authors found factors related to extrinsic motivation (economic benefits, convenience, and perceived usefulness) and intrinsic motivation (sustainability, sense of belonging, trust, and enjoyment) explained 57.7% of the variance in participation intention in the bicycle-sharing program [29]. Ryan et al. (2003) investigated the motivation factors of farm operators in the River Raisin watershed in southeastern Michigan, U.S., and their adoption of conservation practices along riparian zones [30]. The findings indicated farmers exhibiting a stronger intrinsic motivation were more likely to adopt specific conservation practices. Economic motivation, however, did not play a role in farmers' adoption of specific conservation practices [30].

There does not appear to be a consensus in the environmental literature as to how to quantitatively measure intrinsic and extrinsic motivation. For example, Ryan et al. (2003) conducted a factor analysis on 16 items related to conservation practices, resulting in six motivation categories [30]. Four items were used to measure intrinsic motivation: (1) I would be upset if my activities harmed my land, (2) protecting the environment is important to me, (3) I want to conserve this land for future generations, (4) I am very attached to my land and want to protect it ($\alpha = 0.86$). Li and Wen (2019) measured intrinsic motivation using three items about sustainability (e.g., using bicycle sharing helps save natural resources), three items about sense of belonging (e.g., using bicycle sharing allowed me to be part of a group of like-minded people), three items about trust (e.g., I trust that bicycle sharing will be displayed as expected), and three items about enjoyment (e.g., I think using bicycle sharing is enjoyable) [29]. Attitude has also been used as a measure of intrinsic motivation in previous studies (e.g., [31]). Although all scales were deemed reliable, determining the best measures for intrinsic motivation related to environmental sustainability remains unresolved.

Intrinsic Motivation Inventory

The intrinsic motivation inventory (IMI) was developed by Ryan and Deci to assess the subjective experiences of participants from participation in a specific activity and was tested in several experiments (e.g., [32–37]). There are six subscales in the intrinsic motivation inventory: interest/enjoyment, perceived competence, effort, value/usefulness, felt pressure and tension, and perceived choice. The subscales reported factor loadings above 0.6 and no cross-loadings above 0.4. For reference, items that have standardized factor loadings below 0.50 should be removed from the model [38] and cross-loading items are considered above 0.3 in more than one observed factor [39]. Researchers often select subscales that are relevant to the topic of exploration for the inclusion, or exclusion, of specific subscales.

Four subscales were examined in the context of water conservation behavior due to their relevance to the specific activity: interest/enjoyment, perceived choice, perceived

competence, and relatedness. The most direct measure of intrinsic motivation is the interest/enjoyment subscale [40]. Interest and inherent pleasure regarding a specific activity are assessed with this subscale. Perceived choice assesses individuals' sense of freely choosing an activity, and perceived competence evaluates their confidence or self-efficacy in effectively performing a task. Relatedness examines feelings of connection with others [40].

Examples of previous research include McAuley et al. (1989), who investigated four subscales of the IMI (interest/enjoyment, perceived competence, effort, and felt pressure and tension) in the context of physical activity to validate the hypothetical factor structure [41]. The initial model included 18 items. Item deletion was assessed using coefficient alpha, resulting in one item being removed from the interest/enjoyment scale ("While playing this basketball game, I was thinking about how much I enjoyed it") and one item removed from the perceived competence scale ("I am satisfied with my performance at this game"), resulting in a 16-item inventory. Confirmatory factor analysis (CFA) was conducted in two phases to assess the two models using maximum likelihood estimation on the Likert-type scale responses. The CFA results indicated that the 16-item model had stronger internal consistency than the 18-item model, and the authors encouraged additional validation of the measures in sports and social sciences [41].

More recently, Tiago et al. (2017) used the IMI to measure intrinsic motivation in relation to citizen science engagement in Portugal [42]. Seven intrinsic motivation categories were used: interest/enjoyment, perceived competence, effort/importance, perceived choice, value/usefulness, project relatedness, and group relatedness. Statements, adopted from Fonseca and Brito (2001), for each category were rated on a seven-point Likert-type scale, but reliability was not reported. Walker et al. (2018) used the IMI to study the influence of experiential learning on student motivation in a sustainable agriculture tour class [43]. Pre- and post-tour surveys were conducted using the IMI with a seven-point Likert-type scale. All intrinsic motivation inventory constructs were correlated with each other ($p < 0.05$), except for two, which supports the validity of the IMI [43]. The existing body of research supports the utility of the IMI but it has not yet been assessed in the context of water conservation.

The purpose of this study was to determine if the IMI could be adapted in the context of water conservation. The study addresses a gap in the literature by examining an intrinsic motivation scale that can be used in the context of water conservation education, as there does not appear to be a commonly accepted scale. The following research objectives guided the study:

1. Determine and confirm the factor structures associated with respondents' self-reported intrinsic motivation, competence, autonomy, and relatedness towards water conservation.

2. Materials and Methods

2.1. Participants and Procedure

An online questionnaire was used to collect data in June 2022 from residents of Florida, Georgia, and Alabama aged 18 years or older. Respondents were recruited using non-probability opt-in sampling methods, specifically with quota sampling, which is commonly used in public opinion research [44]. Quotas were set for gender, age, and race/ethnicity based on the U.S. Census of the target states. Qualtrics, an online survey platform, was used to recruit respondents. Qualtrics has a large network of participants who have previously agreed to be contacted to participate in research studies [45]. The participants were recruited through a variety of methods, such as membership referrals, ads and promotions, and social networks [45].

The target states for the study were selected due to the ongoing legal battles between the states over shared river basins, which have created an important opportunity to improve participation in water conservation. The dispute originated because of the increasing water consumption in the Atlanta metro area needed for the growing population, which caused harmful environmental consequences in Florida and Alabama [46]. Most recently, Georgia

and Florida have disputed over equitable apportionment of water from the basins; however, the supreme court voted in favor of Georgia [44,47].

A panel of experts in natural resource conservation, survey design, and communication studies reviewed the survey for content accuracy and face validity. The University of Georgia Institutional Review Board (IRB #00005553) approved the design for this research. Informed consent was collected prior to respondents participating in the survey. Potential respondents were provided a short description of the survey content and details that aligned with IRB requirements to help them make an informed decision on if they should participate in the survey. Respondents were asked to click “yes” if they agreed to participate in the research.

The Cochran (1963) guidance was used to determine the sample size needed for this study [48]. The following equation was used:

$$n_0 = \frac{Z^2 pq}{e^2} = \frac{1.96^2 \times 0.5(1 - 0.5)}{0.035^2} = 784 \quad (1)$$

In the equation, sample size is represented with n_0 , the standard distribution of the confidence interval is represented with Z , $1 - p$ is represented with q , and the margin of error is represented with e . A confidence interval of 95%, standard deviation of 0.5, and margin of error of $\pm 3.5\%$ was used for the sample. The maximum variability in a population is accounted for, with a standard deviation of 0.5 [49].

2.2. Measures

Respondents’ self-reported interest/enjoyment (intrinsic motivation) was measured using seven Likert-type scale items on a five-point scale (1 = Strongly Disagree; 2 = Disagree; 3 = Neither Agree nor Disagree; 4 = Agree; 5 = Strongly Agree). The items were adapted from the IMI (see <https://selfdeterminationtheory.org/intrinsic-motivation-inventory/>, accessed on 15 January 2022). Respondents were asked to indicate their level of agreement or disagreement with each of the following statements: (1) I enjoy conserving water, (2) conserving water is fun, (3) conserving water is a boring task, (4) I do not pay attention to conserving water, (5) conserving water is very interesting, (6) conserving is quite enjoyable, (7) while conserving water, I think about how much I enjoy it. See Appendix A for a full list of items included and their normality assumptions.

Respondents’ self-reported perceived competence in water conservation was measured with six Likert-type scale items on a five-point scale (1 = Strongly Disagree; 2 = Disagree; 3 = Neither Agree nor Disagree; 4 = Agree; 5 = Strongly Agree). The items were adapted from the IMI. Respondents were asked to indicate their level of agreement or disagreement with each of the following statements: (1) I think I am pretty good at conserving water, (2) I think I do a good job at conserving compared to others, (3) I feel pretty competent in my ability to conserve, (4) I am satisfied with my ability to conserve water, (5) I am pretty skilled at conserving water, (6) conserving water is a task that I can’t do very well.

Respondents’ self-reported perceived choice (autonomy) in water conservation was measured with seven Likert-type scale items on a five-point scale (1 = Strongly Disagree; 2 = Disagree; 3 = Neither Agree nor Disagree; 4 = Agree; 5 = Strongly Agree). The items were adapted from the IMI. Respondents were asked to indicate their level of agreement or disagreement with each of the following statements: (1) I believe I have some choice in conserving water, (2) I feel like it is not my own choice to conserve water, (3) I do not really have a choice in conserving water, (4) I feel like I have to conserve water, (5) I conserve water because I have no choice, (6) I conserve water because I want to, (7) I conserve water because I have to.

Respondents’ self-reported relatedness to others who conserve water was measured with six Likert-type scale items on a five-point scale (1 = Strongly Disagree; 2 = Disagree; 3 = Neither Agree nor Disagree; 4 = Agree; 5 = Strongly Agree). The items were adapted from the IMI. Respondents were asked to indicate their level of agreement or disagreement with each of the following statements: (1) I feel really distant from people who conserve

water, (2) I really doubt that someone who conserves water and I would ever be friends, (3) I feel like I can really trust people who conserve water, (4) I would like a chance to interact with people who conserve water more often, (5) I would really prefer not to interact with people who conserve water in the future, (6) I don't feel like I can really trust people who conserve water.

2.3. Demographics

Demographics of respondents are reported in Table 1. Respondents represented the three states in similar proportions: Florida (34.1%), Georgia (34.3%), and Alabama (31.6%). Respondents were predominantly White (76.5%), with Alabama having the highest proportion of respondents who identified as White (87.1%). There are notable differences in respondents who identified as Hispanic in Florida (35.9%), as compared to Georgia (8%) or Florida (5.9%). In addition, most respondents were over 55 years in age, which was consistent across all three states (44% in Florida, 46.3% in Georgia, and 46.7% in Alabama). The smallest age group across all three states was respondents aged 18–34.

Table 1. Demographics of respondents (N = 907).

Baseline characteristic	Total (N = 907)		Florida (n = 309)		Georgia (n = 311)		Alabama (n = 287)	
	F	%	F	%	F	%	F	%
Sex								
Female	483	53.3	194	62.8	133	42.8	156	54.4
Male	424	46.7	115	37.2	178	57.2	131	45.6
Age								
18–34 years	206	22.6	81	26.2	66	21.2	59	20.6
35–54 years	287	31.6	92	29.8	101	32.5	94	32.8
55+ years	414	45.8	136	44.0	144	46.3	134	46.7
Race *								
White	694	76.5	212	68.6	232	74.6	250	87.1
Black	130	14.3	56	18.1	49	15.8	25	8.7
Asian	55	6.1	30	9.7	20	6.4	5	1.7
American Indian or Alaska Native	21	2.3	10	3.2	3	1.0	8	2.8
Other	42	4.6	23	7.4	12	3.9	7	2.4
Ethnicity								
Hispanic	153	16.9	111	35.9	25	8.0	17	5.9
Non-Hispanic	754	83.1	198	64.1	286	92.0	270	94.1
Family Income Level								
Less than USD 24,999	195	21.5	62	20.1	58	18.6	75	26.1
USD 25,000–USD 49,999	237	26.1	86	27.8	71	22.8	80	27.9
USD 50,000–USD 74,999	191	21.1	67	21.7	61	19.6	63	22.0
USD 75,000–USD 149,999	218	24.0	77	24.9	85	27.3	56	19.5
USD 150,000–USD 249,999	48	5.3	10	3.2	28	9.0	10	3.5
USD 250,000 or more	18	2.0	7	2.3	8	2.6	3	1.0

Note. * Respondents could select more than one race.

2.4. Data Analysis

The guidance in Green and Salkind (2016) and Costello and Osborne (2005) was used to conduct an exploratory factor analysis (EFA) on the dataset to address the first research objective by determining factor structures from the items using RStudio and the psych and EFAutilities packages [39,50]. Split-data were used for the EFA and CFA for cross-validation to ensure model robustness by determining if the factor structures remain consistent across the data subsets. EFA was conducted on 454 respondents [51] using maximum likelihood estimation [52] and Geomin rotation, which is an oblique rotation that allows the factors to correlate. Oblique rotation was used rather than orthogonal rotation because the correlations among the factors was above 0.32 in some instances [53]. Factor structures were determined using the eigenvalues for a sample correlation matrix greater than 1 and the scree plot.

Bartlett's Test of Sphericity and the Kaiser–Meyer–Olkin Test of Sampling Adequacy were used to assess the suitability of the factor analysis for the data [51]. Bartlett's Test of Sphericity, which determines if the correlation matrix is an identity matrix, should be significant. The KMO statistic, which measures items' shared variance, ranges from 0 to 1, with distinct and reliable variables being closer to 1. A value of 0.8 or higher was used in this study [51].

The study of Costello and Osborne (2005) was used to interpret the EFA factor structures [50]. High communality is regarded to be above 0.8 but in social sciences it is common for the communality to be from 0.4 to 0.7. Items with communalities less than 0.4 need to be carefully considered by the researcher as they may not have a relationship with other items, or additional factor structures may need to be examined. Items that load 0.32 or higher on multiple items may be cross-loading items and may need to be removed from the analysis. Strong loaders are considered to be above 0.50 and are desirable [39].

Confirmatory factor analysis (CFA) was conducted on 453 respondents in MPlus to address the second research objective by confirming factor structures from the EFA results (Mplus Version 8.8, Muthén & Muthén, Los Angeles, CA, USA). Items were removed from the model if they had standardized factor loadings below 0.50. Model fit index criteria were TLI and CFI > 0.90, RMSEA < 0.07 [54,55]. The hypothesized model is represented in the working path diagram in Figure 1. Each proposed latent component covaries with the other latent components. The equation representing the proposed model is as follows:

$$y = \Lambda_y \eta + \varepsilon \quad (2)$$

where y is the observed variable, Λ is the loadings of the latent variables on the observed variables, η is the latent factors, and ε represents the residuals.

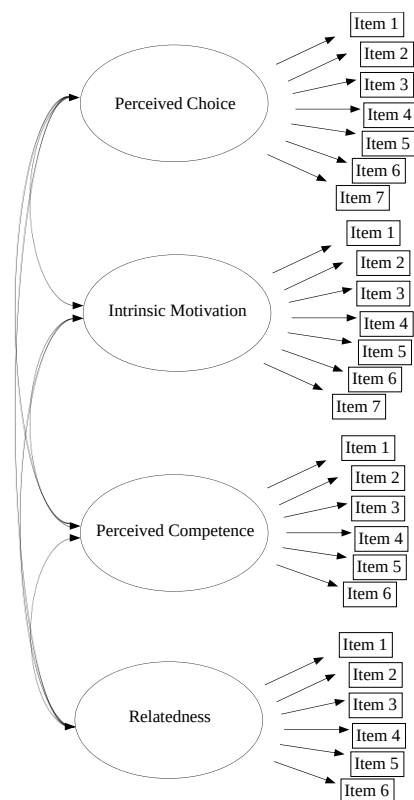


Figure 1. Proposed hypothetical IMI model in the context of water conservation.

3. Results

3.1. Determine the Factor Structures Associated with Respondents' Self-Reported Intrinsic Motivation, Competence, Autonomy, and Relatedness

The first rotation resulted in six distinct components based on the eigenvalues for the sample correlation matrix and the scree plot. Items that loaded 0.32 or higher on multiple items, indicating an issue with cross-loading, were removed from the EFA. They included the following: I feel like it is not my own choice to conserve water; I feel like I have to conserve water; I do not pay attention to conserving water; and I would like a chance to interact with people who conserve water more often. Items that were removed from the EFA because they loaded below 0.5, indicating they were not strong loaders, included: Conserving water is a boring task and I feel like I can really trust people who conserve water.

The second rotation resulted in four distinct components based on the eigenvalues for the sample correlation matrix and the scree plot. Items that were removed from the EFA because they loaded below 0.5, indicating they were not strong loaders, included: I believe I have some choice in conserving water; I conserve water because I want to; and Conserving water is a task that I can't do very well.

The third rotation resulted in four distinct components based on the eigenvalues for the sample correlation matrix and the scree plot (see Table 2). No additional items were removed from the EFA. Table 2 presents the four main factors that emerged from the EFA, including perceived choice (component 1), intrinsic motivation (component 2), perceived competence (component 3), and relatedness (component 4). The factors are each composed of several items that align with the constructs that were theorized in the original IMI. The resulting components were used in the CFA.

Table 2. EFA results from the third rotation that resulted in four components.

Constructs	Item Description	Component 1	Component 2	Component 3	Component 4
Perceived Choice					
	I do not really have a choice in conserving water (RC)	0.565 *	0.005	0.187 *	0.075 *
	I conserve water because I have no choice (RC)	0.886 *	0.03	−0.006	0.042 *
	I conserve water because I have to (RC)	0.679 *	−0.124 *	−0.033	−0.065 *
Intrinsic Motivation					
	I enjoy conserving water	−0.01	0.625 *	0.231 *	0.122 *
	Conserving water is fun	0.043	0.844 *	0.012	−0.018
	Conserving water is very interesting	−0.04	0.771 *	0.029	0.005
	Conserving water is quite enjoyable	0.031	0.920 *	−0.033	−0.009
	While conserving water, I think about how much I enjoy it.	−0.082 *	0.722 *	0.009	−0.122 *
Perceived Competence					
	I think I am pretty good at conserving water	−0.032	0.072 *	0.768 *	0.027
	I think I do a good job at conserving water compared to others	−0.093 *	0.055	0.729 *	0.037
	I feel pretty competent in my ability to conserve	0.028	−0.045	0.827 *	0.02
	I am satisfied with my ability to conserve water	0.066 *	−0.041	0.830 *	−0.056 *
	I am pretty skilled at conserving water	−0.005	0.062 *	0.779 *	−0.066 *
Relatedness					
	I feel really distant from people who conserve water (RC)	0.046	−0.017	0.076 *	0.756 *
	I really doubt that someone who conserves water and I would ever be friends (RC)	−0.03	−0.068 *	−0.004	0.888 *
	I would really prefer not to interact with people who conserve water in the future (RC)	0.036	0.067	−0.086 *	0.620 *
	I don't feel like I can really trust people who conserve water (RC)	0.021	0.041	−0.003	0.599 *

Note. (RC) indicates the items was reverse coded. *. Colored text was used to help distinguish which factor loadings grouped together.

3.2. Confirming the Factor Structures Associated with Respondents' Self-Reported Intrinsic Motivation, Competence, Autonomy, and Relatedness

Model fit indices were examined with the comparative fit index (CFI = 0.94), the Tucker–Lewis Index (TLI = 0.92), and the root mean square error of approximation (RMSEA = 0.07). Factor loadings in the CFA were all above 0.50 [38]; therefore, no items were removed from the model (see Table 3 and Figure 2). Modification indices were examined but not incorporated as the model had adequate fit and adhered to theoretical underpinnings.

Convergent validity was assessed to examine the correlations of the measured constructs on a scale with one another [38]. Convergent validity was satisfied as the average variance extracted (AVE) was above 0.50 for all scales (Table 3). Discriminant validity was assessed to examine if the constructs that should be theoretically distinct from another construct are in fact theoretically distinct. The Fornell and Larcker criterion was used to assess discriminant validity in the model based on factor correlations less than the square root of their corresponding AVE [56]. Discriminant validity was satisfied (Table 4). The Cronbach's alpha for the final constructs indicated satisfactory internal consistency (Table 3).

Table 3. CFA results for the adapted IMI model in the context of water conservation.

Construct	Item Description	Estimate (Standardized)	Cronbach alpha (α)	Average Variance Extracted	Composite Reliability
Perceived Choice		-	0.75	0.54	0.77
	I do not really have a choice in conserving water (RC)	0.585			
	I conserve water because I have no choice (RC)	0.913			
	I conserve water because I have to (RC)	0.669			
Intrinsic Motivation		-	0.90	0.65	0.90
	I enjoy conserving water	0.762			
	Conserving water is fun	0.839			
	Conserving water is very interesting	0.801			
	Conserving water is quite enjoyable	0.883			
	While conserving water, I think about how much I enjoy it.	0.754			
Perceived Competence		-	0.90	0.64	0.90
	I think I am pretty good at conserving water	0.821			
	I think I do a good job at conserving water compared to others	0.776			
	I feel pretty competent in my ability to conserve	0.793			
	I am satisfied with my ability to conserve water	0.785			
	I am pretty skilled at conserving water	0.81			
Relatedness		-	0.82	0.54	0.82
	I feel really distant from people who conserve water (RC)	0.796			
	I really doubt that someone who conserves water and I would ever be friends (RC)	0.868			
	I would really prefer not to interact with people who conserve water in the future (RC)	0.623			
	I don't feel like I can really trust people who conserve water (RC)	0.605			

Note. (RC) indicates the items was reverse coded.

Table 4. Discriminant validity of constructs based on the Fornell and Larcker criterion.

Construct	Perceived Choice	Intrinsic Motivation	Perceived Competence	Relatedness
Perceived Choice	0.74	-	-	-
Intrinsic Motivation	−0.30	0.81	-	-
Perceived Competence	−0.05	0.61	0.8	-
Relatedness	0.51	−0.06	0.14	0.73

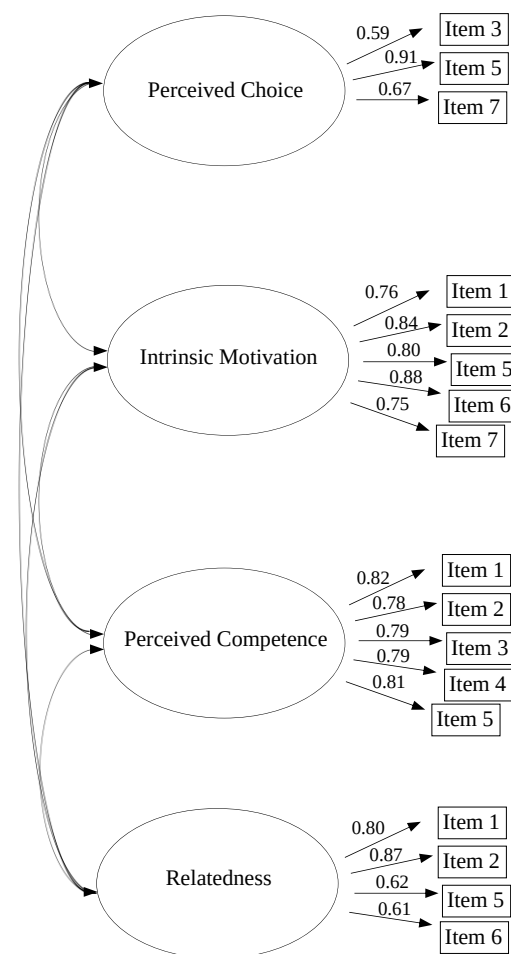


Figure 2. Adapted IMI model in the context of water conservation.

4. Discussion

There are two important practical contributions from the study for water conservation research and education. First, the study results indicate that the adapted CFA model, hereby referred to as the Water Conservation IMI model, has sufficient internal consistency and is therefore an empirically validated measure of intrinsic motivation regarding residential water conservation behavior. Second, measures for intrinsic motivation are likely to be context specific, and researchers should determine if the proposed model is applicable to environmental contexts beyond water conservation, such as other UN Sustainable Development Goals including life on land. These contributions are detailed in the following paragraphs.

The results indicate the Water Conservation IMI model may be used as a measure for intrinsic motivation in studies about individual water conservation behavior. Four items were removed from perceived choice, two items were removed from intrinsic motivation, one item was removed from perceived competence, and two items were removed from relatedness. Previous research on the IMI suggests that items may need to be removed depending on the context of the study; therefore, this finding is consistent with the intention of the scale (see <https://selfdeterminationtheory.org/intrinsic-motivation-inventory/>, accessed on 15 January 2022). In addition, the EFA and CFA models resulted in items loading onto the components/ constructs that corresponded with their loading from the initial IMI. For example, the items “I do not really have a choice in conserving water”, “I conserve water because I have no choice”, and “I conserve water because I have to” loaded onto the perceived choice component for the intrinsic motivation inventory and in the present study. The development of this adapted inventory may provide an opportunity for water sustainability education to have a greater impact on water consumption. Future studies should examine if the adapted IMI components are helpful predictors of water conservation behaviors to

determine the usefulness of the scale for practical implications. For example, adding intrinsic motivation to the theory of planned behavior may help explain the additional variance in behavior that is often called into question as the intention–behavior gap [57].

The four components of the IMI had all initially included items that were positively worded and items that were negatively worded. The items that were negatively worded needed to be reverse coded prior to analysis. The final components from the EFA either had items that were all positively worded or all negatively worded, and there was not a component that included both positively worded and negatively worded items. However, not all items that remained in the final model were those using a specific frame of negative or positive. For example, perceived choice initially had seven items. The first EFA rotation removed two items that were negatively worded because they were cross-loading items. The second rotation removed the two items that were positively worded because they were not strong loaders. The remaining three items were all negatively worded. Future studies may want to consider using items for constructs that are either negatively or positively worded, perhaps in the context of ordinal versus continuous data.

The adapted IMI items asked respondents about their individual water conservation behavior. However, introducing the context of the water conservation behavior, such as landscape irrigation or water saving household appliances, may introduce additional nuance into responses to the measures. Civic water-related behavior, such as donating to an organization that protects water, is another important component of water conservation that could be examined in the context of intrinsic motivation [58].

Studies regarding other environmental topics, such as recycling or green consumption, should consider adapting items from the IMI to determine if scale use is consistent and comparable across environmental issues. It is possible, however, that items in the IMI are context-specific in the realm of environmental issues. For example, recycling and water conservation may require different measures for intrinsic motivation, especially since recycling is a tangible behavior and water conservation is not. Another example is reducing meat consumption, as food culture and existing beliefs regarding health may influence intrinsic motivation variables as compared to water conservation [59]. However, having comparable measures would be helpful as educational programs are developed. Therefore, the IMI should be tested in different environmental contexts in future studies. Since conservation educators are hindered by a lack of rich audience data to inform behavioral interventions, and tools to assess these factors themselves, such measures could support more effective consumption reduction. In addition, environmental studies that utilize the IMI generally adapt the items without exploring or confirming their suitability in that context (e.g., [42,43]), causing uncertainty about the validity and reliability of the adapted scales. The limited number of environmental studies that explore and confirm the IMI in a specific context also makes a comparison of the final factors included in the model difficult.

The current study focused on the IMI, which measures facets of intrinsic motivation. However, Li and Wen (2019) found that extrinsic motivation was a stronger predictor of intention to participate in a bicycle-sharing program than intrinsic motivation [29]. Ryan et al. (2003) used economic motivation, which is related to extrinsic motivation, to predict specific conservation practices for farms along riparian zones [30]. Economic motivation was not a significant predictor, but extrinsic motivation is represented by more than economic incentives. Thus, a scale for extrinsic motivation needs to be further explored and validated in the environmental literature to establish a strategy to quantify and compare both intrinsic and extrinsic motivation (e.g., [29,30]).

4.1. Application

The Water Conservation IMI model can be applied in educational contexts by integrating the model into curriculum frameworks. For instance, administering the Water Conservation IMI model to participants at the beginning of a week-long water education camp can provide an overview of their motivations to conserve water and inform educational practices throughout the week. If future studies determine the Water Conservation

IMI model to be a helpful predictor of water conservation behaviors, incorporating motivational techniques from the Water Conservation IMI model may increase engagement in conservation. For example, an adult education program may provide skill-building activities where participants learn to create rainwater collection systems, influencing perceived competence. Another opportunity is to encourage social support by establishing peer groups to work on water conservation projects, influencing perceived competence and relatedness.

4.2. Limitations

There are several study limitations that should be addressed prior to interpretation of the results. First, the items were self-reported measures of intrinsic motivation. Self-reported measures lend to social desirability bias, where respondents answer items in a way that they perceive is socially acceptable [60]. Social desirability bias may lead to over-reporting of traits or behaviors that are perceived as desirable by the survey respondent. In addition, social desirability bias may look different for different cultural groups as cultural norms regarding modesty and exaggeration can cause different interpretations of a survey item. Similarly, self-reported measures may be influenced by respondents' self-awareness, as respondents' insight on their own beliefs or behaviors may be skewed. This is especially relevant regarding water conservation, as it is not a tangible action and is therefore more difficult for respondents to conceptualize unless they are directly engaging with their water bill. There are very few studies that examine if self-reported measures are consistent with actual or observed measures, especially in the context of water conservation, and the significance of the discrepancy in actual versus self-reported measures is therefore unknown.

Next, the respondents were from Florida, Georgia, and Alabama, and therefore are not representative of the broader U.S. It is possible that including other states with distinct water issues would result in different findings. For example, California has very different experiences with and responses to water issues than Florida, Georgia, and Alabama, which may impact residents' motivation towards water conservation. Future studies may want to examine the between- and within-impact difference of state-level variables on water conservation to understand the nuances of geographical location on environmental sustainability and consumption reduction.

Another study limitation was the use of non-probability opt-in sampling [61]. This study specifically used quota sampling, which splits the population into segments and fills a quota when people match certain characteristics for each segment [62,63]. For example, quotas can be based on certain numbers from the census to produce a sample that is a "miniature approximation of the population with respect to the selected characteristics" ([62], p. 169). Probability sampling was not used due to financial constraints and a decline in response rates [61]. In addition, public opinion research data collection has largely shifted to using non-probability opt-in samples due to the listed constraints of probability samples [38]. Regardless, there are several limitations of non-probability opt-in sampling that need to be addressed, such as bias from under coverage and non-response, selection bias, and limited use for causal inference. Bias from under coverage and non-response may occur as respondents may not have internet access [47]. Selection bias may occur because there are specific types of people who participate in online surveys. However, this is also possible in probability sampling where certain people are more likely to answer phone or mail surveys [38]. Limited use for causal inference occurs due to a lack of randomization and control. Quota sampling was used to address some limitations of non-probability opt-in sampling by collecting data representative of the target population based on U.S. Census demographics, which reduces the risk of groups being over- or under-represented [38].

5. Conclusions

Future food security, economic growth, and healthy ecosystems require the sustainable use of water resources by all stakeholders [2]. The current study sought to understand

intrinsic motivation towards water conservation to help inform educational interventions on residential water use. The study findings contribute to the literature by exploring ways to quantify outcomes of sustainability education interventions, helping educators determine effective programming to decrease the excessive consumption of natural resources and achieve the UN Sustainable Development Goals.

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Appendix A

Table A1. Items used to measure latent constructs and their corresponding normality assumptions.

	Mean	SD	Skew	Kurtosis	Included in Final Model
Intrinsic motivation					
I enjoy conserving water	3.82	0.84	−0.36	0.00	Yes
Conserving water is fun	3.41	0.92	0.03	−0.24	Yes
Conserving water is a boring task (RC)	3.36	1.03	−0.21	−0.46	No
I do not pay attention to conserving water (RC)	3.76	1.09	−0.79	0.00	No
Conserving water is very interesting	3.33	0.99	−0.22	−0.10	Yes
Conserving water is quite enjoyable	3.32	0.94	−0.12	−0.03	Yes
While conserving water, I think about how much I enjoy it.	2.98	1.09	0.02	−0.48	Yes
Perceived competence in water conservation					
I think I am pretty good at conserving water	3.81	0.86	−0.80	0.92	Yes
I think I do a good job at conserving water compared to others	3.75	0.85	−0.50	0.40	Yes
I feel pretty competent in my ability to conserve	3.89	0.82	−0.90	1.52	Yes
I am satisfied with my ability to conserve water	3.93	0.79	−0.73	0.92	Yes
I am pretty skilled at conserving water	3.66	0.87	−0.43	0.24	Yes
Conserving water is a task that I can't do very well (RC)	3.58	1.14	−0.56	−0.53	No

Table A1. Cont.

	Mean	SD	Skew	Kurtosis	Included in Final Model
Perceived choice (autonomy) in water conservation					
I believe I have some choice in conserving water	4.30	0.70	−1.19	2.94	No
I feel like it is not my own choice to conserve water (RC)	3.45	1.28	−0.47	−0.95	No
I do not really have a choice in conserving water (RC)	3.86	1.06	−1.02	0.65	Yes
I feel like I have to conserve water (RC)	2.57	1.17	0.41	−0.68	No
I conserve water because I have no choice (RC)	3.71	1.11	−0.78	0.00	Yes
I conserve water because I want to	4.24	0.80	−1.25	2.43	No
I conserve water because I have to (RC)	3.35	1.18	−0.46	−0.62	Yes
Relatedness to others who conserve water.					
I feel really distant from people who conserve water (RC)	3.85	1.05	−0.77	0.09	Yes
I really doubt that someone who conserves water and I would ever be friends (RC)	3.96	1.04	−0.85	0.21	Yes
I feel like I can really trust people who conserve water	3.26	0.90	−0.20	0.56	No
I would like a chance to interact with people who conserve water more often	3.17	0.96	−0.30	0.20	No
I would really prefer not to interact with people who conserve water in the future (RC)	3.61	1.08	−0.38	−0.38	Yes
I don't feel like I can really trust people who conserve water (RC)	3.84	1.02	−0.59	−0.13	Yes

Note. (RC) indicates the items was reverse coded.

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